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A novel scenario-based robust bi-objective optimization model for humanitarian logistics network under risk of disruptions

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ABSTRACT

Humanitarian aid in disasters is critical to saving lives and alleviating human suffering. This paper presents a novel scenario-based robust bi-objective optimization model that integrates medical facility location, casualty transportation, and relief commodity allocation considering triage. The proposed model aims to minimize the total deprivation cost of casualties due to the delayed access to medical services and the total operation cost. Following a set of disruption scenarios, the scenario-based robust approach is applied to protect solutions against the risk of disruptions in temporary medical centers. Considering the uncertain number of casualties under each scenario, the robust method which denotes the uncertainty as interval data is adopted. We utilize the ε-constraint method to deal with the bi-objective model. Additionally, we consider real case studies of the Wenchuan Earthquake to validate the proposed model. Several numerical experiments are conducted to examine the effects of uncertainties and capacities of medical facilities on the main objective value. The performance of considering the uncertainty and facility disruption is also discussed.

1. Introduction

Severe disasters either natural (floods, earthquakes, hurricanes, mudslides, etc.) or human-made (wars, terrorist attacks, etc.) have caused a considerable number of casualties and great economic losses (Rahmani, 2019). For example, the Japan earthquake struck in the western Pacific region on 11 March 2011, causing 19,575 people dead, 6157 injured, 2585 missing, and approximately 235 billion direct economic losses. After natural disasters occurred, there are numerous casualties in the affected areas who need to be transported due to the powerful destructiveness. Besides, the relief commodities in disaster areas are always insufficient to meet all demands. Thus, it is vital to design a humanitarian logistics network (HLN) for transporting casualties and allocating relief commodities.

Most of the existing literature assumes that the medical facilities in the HLN remain available and are not affected by disasters. However, some facilities can be easily disrupted due to secondary disasters, especially in large-scale disasters such as earthquakes. For instance, there were more than 1,000 medical facilities disrupted in the Nepal earthquake, which disturbed the delivery of medical services (Akbarpour et al., 2020). This fact motivates us to carefully consider the risk of facility disruptions when configuring the HLN. There are two popular approaches to designing a reliable HLN in response to facility disruptions. The first method is considering a substitution to avoid the entire network being affected (Jabbarzadeh et al., 2016). In the second method, since the occurrence of disaster is not always predictable, the decision-maker must account for all possible occurrence scenarios and decide the locations of

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facilities before the potential disaster occurs (Kaveh and Ghobadi, 2017). However, finding a substitution often leads to great cost and economic losses (Shen et al., 2011). As discussed by Snyder and Daskin (2006), a robust rescue network should work consistently well in all scenarios. In this paper, we model the facility disruptions using a set of scenarios. And the scenario-based robust method proposed by Mulvey et al. (1995) is utilized to get stable solutions across all disruption scenarios. In the scenario-based robust method, the solution robustness is introduced and measured by the variance among objective values of all scenarios. Compared to the stochastic programming method, the solutions generated by the scenario-based robust method are less sensitive to changes in the data defined by scenarios.

In general, we have no way to accurately predict the severity of the catastrophes. It is very difficult to estimate the exact values of the key parameters in advance such as casualty number. Besides, the mistakes in predicting these values can have serious negative consequences (Jabbarzadeh et al., 2014). As a result, it is necessary to design the HLN that is immunized against the uncertainties. Stochastic optimization is a well-known method to deal with uncertainties. However, it is difficult to estimate the probability distributions of uncertain parameters due to insufficient historical data. If we assume probabilistic distributions for the uncertain parameters, the differences from the real probabilistic distributions may make decisions unsuitable for the actual situations. As another popular approach to cope with uncertainties, the robust optimization method (Bertsimas and Sim, 2004; Ben-Tal et al., 2009) describes the uncertainties as interval data that fluctuates around a nominal value regardless of the probability distribution. Since the historical data is inadequate, it is easier to predict the interval values than point values and the corresponding probabilities. Besides, many approaches have been developed to forecast the intervals of uncertain parameters (Wang et al., 2016; Sun et al., 2018; Zhu et al., 2019a). Thus, we apply the robust optimization approach developed by Ben-Tal et al. (2009) for handling the uncertain number of casualties, which introduces the uncertain budget to control the uncertain level and the conservatism of solutions.

Unlike commercial logistics, emergency rescue operations emphasize alleviating the human suffering caused by the lack of a good or service. Holguín-Veras et al. (2013) firstly defined the economic valuation of human suffering as the “deprivation costs”, which is an external effect associated with rescue activities. They also compared different proxy methods to incorporate human suffering. Then, Holguín-Veras et al. (2016) applied the Contingent Valuation (CV) approach to estimate the Deprivation Cost Functions (DCFs) for drinking water, the findings showed a non-linear relationship between the deprivation cost of drinking water and the duration of deprivation. Most of the existing studies for disaster response planning usually ignore human suffering, only a few researchers consider the deprivation cost in the HLN. These DCFs can be divided into two categories. One is related to the delayed access to medical service (Zhu et al., 2019b), and the other is associated with the delayed access to relief commodities (Loree and Aros-Vera, 2018; Cotes and Cantillo, 2019; Paul and Wang, 2019). Since we consider that the deprivation cost is caused by the lack of medical service, the DCF proposed by Zhu et al. (2019b) is applied and improved in this study. However, Zhu et al. (2019b) only addressed the casualty routing problem considering triage and the deprivation cost. We focus on investigating the joint planning of locating medical facilities, evacuating casualties, and distributing relief commodities with the consideration of deprivation cost and facility disruption.

After an earthquake, there are a considerable number of casualties with different injuries. As different types of casualties require different relief commodities and medical treatments, it is necessary to transport casualties with triage for better treatment. The emergency personnel who firstly arrive at the affected areas usually assess the physical condition of casualties according to the severity of injuries and quickly classify the casualty. There are several commonly used approaches of on-site triage, including Simple Triage Algorithm and Rapid Treatment (START) developed by the hospital and fire department of Newport Beach, Injury Severity Score (ISS) presented by Baker et al. (1974), and Sacco Triage Method (STM) proposed by Sacco et al. (2005). These on-site triage methods classify the casualties based on their important vital signs (consciousness, pulse, respiration, blood pressure, etc.) and injured parts (chest, head, abdomen, neck, spine, spinal cord, etc.). The results of casualty classification are the same regardless of the triage method. Thus, the choice of the triage method does not affect the final decisions and is not the focus of studying.

In this paper, the historical data is considered sufficient to estimate the disruption scenarios, and the data about casualty number is not enough. We apply a set of scenarios to model the facility disruption, and the robust approach presented by Mulvey et al. (1995) is adopted to obtain a stable solution across all disruption scenarios. The uncertainty in the number of casualties under each disruption scenario is addressed by the robust optimization method developed by Ben-Tal et al. (2009). Combining the above-mentioned two robust approaches (Mulvey et al., 1995; Ben-Tal et al., 2009), we develop a novel scenario-based robust bi-objective (NSRB) optimization model to locate medical facilities, transport casualties, and allocate relief commodities considering triage and deprivation cost. The model aims to minimize the total deprivation cost and the total operation cost, then the ε-constraint method is applied to deal with the bi-objective model.

The rest of this paper is presented as follows: Section 2 briefly reviews the related literature, and the main contributions are also highlighted. The problem definition and mathematical model are defined in Section 3. Section 4 demonstrates our model via case studies based on the Wenchuan earthquake, and provides the computational results. Finally, the concluding remarks and the direction for future research are presented in Section 5.

2. Literature review

We review the related literature from three aspects: (1) the integration problems associated with HLN that combine facility location with relief supply allocation or casualty evacuation; (2) HLN design problems based on robust optimization; (3) HLN design problems considering facility disruptions.

2.1. Integration problem of HLN

There are some studies that optimize the decisions about facility location and relief supply allocation from the relief aspect, because the relationship between the two decisions is inextricable. Mete and Zabinsky (2010) studied the joint planning of facility location and resource allocation by minimizing the total cost and the total transportation time. Stochastic programming was utilized to cope with the uncertainties in demand and travel time. Liu et al. (2016) proposed a two-stage optimization framework to locate warehouses and store materials for pollution accidents under uncertainties. The objectives included the minimization of total cost and total distance between the supportive center and the demand point. Tofighi et al. (2016) presented a novel two-stage scenario-based possibilistic-stochastic programming model for total operation cost minimization. And the cost, transportation time, demand, and capacity were considered to be uncertain. Wang et al. (2021) presented a stochastic programming model to minimize the total cost considering the uncertain number of affected people and traffic congestion. Two solution approaches based on particle swarm optimization and variable neighborhood search were introduced for large-sized problems. After disasters occurred, the relevant information may change over time. Therefore, the dynamic problems have been investigated in some studies. Moreno et al. (2018) presented a two-stage multi-period stochastic programming model for disaster response under uncertainties about demand and supply, which aimed to minimize the expected cost. Baharmand et al. (2019) developed a multi-period MIP model for disaster response, which considered multiple layers of disaster zones and imposed constraints on the number of facilities and vehicles. The objectives consisted of the minimization of cost and the response time.

Besides, some scholars have addressed the integration problem of facility location and casualty evacuation from the evacuation aspect. In some literature, the parameters were considered to be deterministic. Salman and Gül (2014) presented a MIP model with minimizing the cost of facility construction, and the sum of transportation and waiting time for wounded, taking the extra service capacity into account. Caunhye et al. (2015) studied the facility location and victim allocation decisions with resource limitations and triage for radiological events. The model aimed at minimizing the weighted sum of travel time. Liu et al. (2019) proposed a bi-objective model to maximize the expected survivors and minimize the total cost, which accounted for casualty triage, survival probability, and medical resources constraints. An iteration method was developed for the Pareto frontier. In recent years, the integration problems that optimized location and evacuation decisions under uncertain environments have been gaining prestige. Caunhye and Nie (2018) applied the stochastic programming method to address uncertainties about demand and travel time considering triage and self-evacuees. They introduced an algorithm on the basis of Benders decomposition for faster computation. Hu et al. (2019) presented a MIP model that integrated the decisions of locating shelters and shifting the injured with consideration of the diurnal population shifts. The uncertainty in the number of injured people was captured by a set of scenarios.

In addition to focusing on the relief aspect or evacuation aspect, several studies also integrated the two aspects in a mathematical model. However, few studies accounted for human suffering in their models. Setiawan et al. (2019) proposed three models for relief allocation and victim evacuation. In the first model, the two kinds of operations were separated. In the second model, relief can be allocated across administrative boundaries. The last model allowed the two kinds of operations to share vehicles. Ghasemi et al. (2020) developed a stochastic programming model that optimized both relief supply distribution and victim evacuation. The complicated network was utilized to model the uncertain demand and chance constraint programming method was applied to obtain the deterministic model of equivalence. Haeri et al. (2020) developed a bi-level model to pre-position relief items, distribute relief items, and transfer the injured in preparedness and response stages. The model aimed to simultaneously minimize the unsatisfied demand and the expected cost. They improved the data envelopment analysis approach to locate relief centers, and a novel fuzzy approach was introduced to handle the uncertain demand for relief items and the uncertain number of the injured. From the reviewed literature, we can find that the integration problem of facility location, supply allocation, and casualty evacuation has been relatively less studied.

2.2. HLN design problems based on robust optimization

As highlighted by Starr and Van Wassenhove (2014), achieving robust solutions should be regarded as the main goal in emergency logistics due to the data deficiency. Some studies have employed the interval robust optimization method proposed by Bertsimas and Sim (2004) or Ben-Tal et al. (2009) to address the uncertainty, which defined the uncertain parameters as distribution-free interval data. The uncertain budget is introduced to control the conservatism of the solution. Zhang and Jiang (2014) presented a bi-objective robust model for delivering medical services under uncertain demand. The model aimed to minimize the total cost and maximize the area coverage. Li and Chung (2019) addressed two vehicle routing problems to deliver critical commodities by assuming whether the vehicle capacity is sufficient to meet all demands of each customer or not, and applied the robust optimization method to handle the uncertainties in transportation time and demand. A two-stage heuristic approach was developed to deal with large-sized instances. Wang and Paul (2020) presented a robust optimization model for the facility location and supply allocation in hurricane preparedness considering the uncertain cost and casualty number, which aimed to minimize the sum of deprivation cost and operation cost (i.e., the total social cost). Considering the uncertain demand and travel time, Ahmadi et al. (2020) developed a two-stage robust model in response to disasters. In the first stage, they attempted to allocate supplies effectively by maximizing the demand coverage. In the second stage, the routing of delivering supplies was optimized by minimizing the total completion time. Sun et al. (2021) brought up a 0–1 integer programming model to determine the facility location, resource distribution strategies, and victim evacuation plans. They focused on minimizing the total Injury Severity Score and the total cost. The interval robust optimization method was also utilized to handle the uncertainty in casualty number, travel time, and demand.

As another popular robust method, the scenario-based robust approach proposed by Mulvey et al. (1995) measures the performance of a solution across a set of scenarios by introducing the solution robustness and the model robustness. Rezaei-Malek et al. (2016)

developed a bi-objective scenario-based robust model to locate warehouses and allocate perishable commodities considering uncertainties of demand, cost, and travel time. The two objectives included the minimization of average response time and the total cost. Haghi et al. (2017) proposed a scenario-based robust model that integrated facility location, relief commodity distribution, and casualty evacuation considering uncertain demand and cost. The MOGASA algorithm was developed to solve the large-scale problem. Habibi-Kouchaksaraei et al. (2018) addressed the multi-period blood supply chain design problem under uncertain cost, demand, and capacity via formulating a bi-objective scenario-based robust model, which consisted of minimizing the cost and the blood shortage. Safaei et al. (2018) presented a bi-level mathematical model to distribute relief supplies considering the minimization of the total cost, and the scenario-based robust method was adopted to handle the uncertain demand. Haghjoo et al. (2020) proposed a scenario-based robust model to design a dynamic blood supply chain network under uncertainties in demand and disruption probability. All of the above-mentioned studies considered that the parameters under each scenario were deterministic. However, few studies have combined the two robust methods to describe the uncertainties in each disaster scenario.

2.3. HLN design problems considering facility disruptions

Although there is substantial literature relevant to HLN design problems, the disruption of facilities is less investigated. Ransikarbum and Mason (2016a) addressed the integration problem of emergency supply distribution and network restoration considering both node and arc disruptions. They developed a three-objective model that incorporated equity as a criterion to support the emergency supply distribution. Moreover, the loss data was estimated by a geographic information system-based software package. Ransikarbum and Mason (2016b) further proposed a goal programming-based multi-objective model to optimize the strategic decisions of relief supply distribution and network restoration, and the partial restoration was taken into account. Diabat et al. (2019) considered facility location and blood distribution in a robust model by combing the maximum regret and expected value for all scenarios in the

Table 1
Comparison between the related literature and this paper.

Reference	Facility location	Casualty evacuation	Supply allocation	Deprivation cost	Facility disruption	Uncertainty	Uncertainty approach	Objective functions
Mete and Zabinsky (2010)	√		√			Demand; Travel time	Stochastic optimization	Total cost; Total travel time
Zhang and Jiang (2014)	√		√			Demand	Interval robust optimization	Total cost; Area coverage
Ransikarbum and Mason (2016a)	√		√		√			Equity; Total unsatisfied demand; Total cost
Rezaei-Malek et al. (2016)	√		√			Demand; Cost; Travel time	Scenario-based robust optimization	Average response time; Total cost
Haghi et al. (2017)	√	√	√			Demand; Cost	Scenario-based robust optimization	Total cost; Satisfaction level
Safaei et al. (2018)	√		√			Demand	Scenario-based robust optimization	Total cost
Caunhye and Nie (2018)	√	√				Demand; Travel time	Stochastic optimization	Total travel time
Rahmani (2019)	√		√		√	Demand; Cost	P-criterion approach; Interval robust optimization	Total cost
Yahyaei and Bozorgi-Amiri (2019)	√	√	√		√	Number of affected people	Interval robust optimization	Total cost
Haeri et al. (2020)	√	√	√			Demand	Fuzzy method	Unsatisfied demand; Expected cost
Haghjoo et al. (2020)	√		√		√	Demand; Disruption probability	Scenario-based robust optimization	Total cost
Wang and Paul (2020)	√	√	√	√		Cost; Casualty number	Interval robust optimization	Total social cost
Hamdan and Diabat (2020)	√		√		√	Demand; Supply	Stochastic optimization	Response time; Total cost
Mohammadi et al. (2020)	√	√	√		√	Demand; Cost; Travel time; Capacity	Interval robust optimization; Neutrosophic set	Total cost; Total time; Cost variation
This study	√	√	√	√	√	Casualty number	Interval robust optimization; Scenario-based robust optimization	Deprivation cost; Operation cost

objective function, which accounted for the uncertain demand along with the disruptions of facilities and routes. Rahmani (2019) studied the dynamic blood supply chain design problem by minimizing the operation cost. The p-criterion approach was applied to protect the solutions from the risk of facility disruption, and the interval robust method was adopted to cope with the uncertainties in demand and some cost parameters. Hamdan and Diabat (2020) brought up a two-stage stochastic optimization model to design an emergency blood network, which aimed at minimizing the response time and the total cost. They used several scenarios to describe facility disruption, route disruption, uncertain demand, and uncertain blood supply. Ransikarbum and Mason (2021) proposed a bi-objective optimization model for post-disaster supply distribution and network restoration. The hybrid NSGA-II algorithm was developed to solve the bi-objective model.

Apart from the above-mentioned studies that considered the risk of facility disruption in the relief supply allocation problem, some scholars have also considered the facility disruption in the integration problem of facility location, relief supply allocation, and casualty transportation. Yahyaei and Bozorgi-Amiri (2019) developed a MIP model that optimally located facilities, allocated relief goods, and evacuated people considering the disruption probability of distribution centers. The robust method was used to tackle the uncertain number of people, which denoted the uncertain parameter as interval data. Mohammadi et al. (2020) presented a robust neutrosophic model to site facilities, distribute victims, and allocate relief commodities with trucks under uncertainties about demand, cost, travel time, and capacity. The disruption probability of the distribution center was also considered in the location-routing problem. The reviewed literature all considered that the distribution center related to allocating relief supplies was at the risk of disruption, and most of them only focused on the allocation of relief supplies. However, we investigate the joint planning of facility location, relief supply allocation, and casualty evacuation considering the disruption of medical facilities.

According to the research reviewed above, few studies integrated relief aspect and evacuation aspect considering triage, deprivation cost, facility disruptions, and uncertainties in the HLN. To further clarify the research gap and the contributions of this study, Table 1 offers a comparative literature review by listing the related studies on the emergency rescue problem according to their main features. The main contributions of this paper can be concluded as follows:

(1) We integrate the facility location, casualty transportation, and relief commodity allocation in emergency rescue campaigns. The casualties are classified into two categories, and the deprivation cost with respect to the lack of medical treatment is considered. Besides, we present different functions to describe the deprivation cost of the two classes of casualties over time;
(2) The disruption risk of medical facilities associated with disaster scenarios is considered. The scenario-based robust approach is applied to get the stable solutions across all disruption scenarios;
(3) We apply the interval robust approach to address the uncertain casualty number in each scenario, and develop the NSRB model by incorporating the interval robust method and the scenario-based robust method. The objectives consist of minimizing the total deprivation cost and the total operation cost. The penalty cost of un-transferred casualties is imposed on the objective to get more realistic solutions;

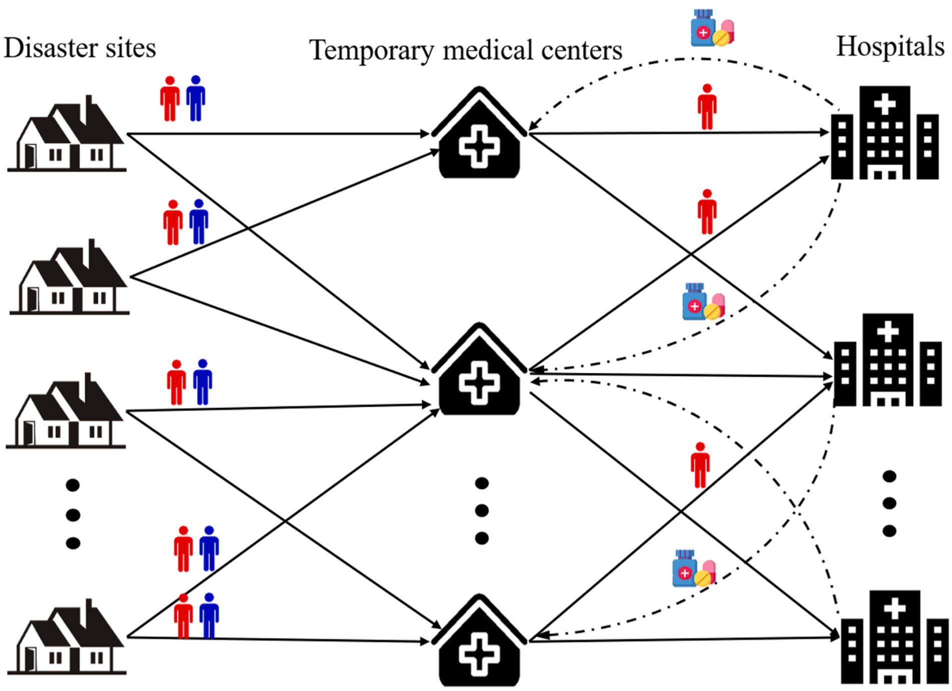


Fig. 1. Structure of the proposed humanitarian logistics network.

(4) We use the real case of the Wenchuan earthquake to illustrate the validity and robustness of our proposed modeling approach. The impacts of uncertain casualty number and the medical facility capacity are analyzed. Besides, the effects of considering the uncertainty and facility disruption are evaluated.

3. Problem definition and formulation

We establish a three-echelon HLN that consists of disaster sites, temporary medical centers, and hospitals. Following the catastrophic earthquake, there are plenty of casualties at disaster sites. The emergency practitioners who firstly reach the disaster sites classify the casualties into two groups based on their vital signs and injured parts: mild and serious casualty. In order to improve relief effectiveness, facilities closer to the disaster sites should be selected as candidate temporary medical centers, and the candidate hospitals are far from the disaster sites. Mild casualties are directly transported to temporary medical centers for treatment. Due to the severity of injuries, serious casualties are first transferred to temporary medical centers to relieve injuries. Then serious casualties are sent to hospitals for medical treatment. Besides, hospitals could provide relief commodities for casualties at temporary medical centers. The structure of our HLN is described in Fig. 1.

Temporary facilities are at risk of disruption because of the proximity to the disaster area. We consider various scenarios associated with the disruptions of temporary medical centers to get a more realistic solution. Because it is very difficult to accurately estimate the information about the number of casualties, the casualty number under each scenario is assumed to be uncertain. The interval robust method is applied to deal with uncertainty about the casualty number. Unlike Zhu et al. (2019b), the deprivation cost functions of the two types of casualties are different. More specifically, the deprivation cost function for mild casualties is not three-stage, because they are transported to temporary medical centers for complete treatments. The deprivation cost of all casualties increases exponentially as the waiting time increases before arriving at the temporary medical centers for services. During receiving rapid medical treatment at the temporary medical centers, the deprivation cost of serious casualties tends to decrease linearly. After being treated at the temporary medical centers, the deprivation cost of serious casualties begins to show a new slightly slower exponential increase. The probability density functions of the deprivation cost for the mild casualty and serious casualty are defined as

$$f(t) = e^{g_1 t} + e^{h_1}, 0 < t \leq tc_{ij} \quad \text{and} \quad f^{'}(t) = \begin{cases} e^{g_2 t} + e^{h_2}, 0 < t \leq tc_{ij} \\ -g_3 t + h_3, tc_{ij} < t \leq tc_{ij} + ct_j \\ e^{g_4 t} + e^{h_4}, tc_{ij} + ct_j < t \leq tc_{ij} + ct + th_{jh} \end{cases} \quad \forall i \in I, j \in J, j \in H$$

respectively, where tc_{ij} is the travel time between disaster site i and temporary medical center j, ct_j denotes the rapid treatment time of serious casualties at the temporary medical center j, th_{jh} represents the transportation time of serious casualties from temporary medical center j to hospital h. Fig. 2 shows the changes in probability density functions over time. The deprivation cost of the casualty is calculated using the definite integral.

In this problem, the decisions include (1) the location of temporary medical centers and hospitals; (2) the flow of mild casualties from disaster sites to temporary medical centers; (3) the flow of serious casualties from disaster sites to hospitals via temporary medical centers; (4) the flow of relief commodities from hospitals to temporary medical centers. Considering the human suffering caused by the unavailability of medical service, we aim to minimize the total deprivation cost for two types of casualties, and minimize the total operation cost that consists of the construction cost for temporary medical centers, transportation cost for casualties, and transportation cost for relief commodities. The main assumptions are as follows:

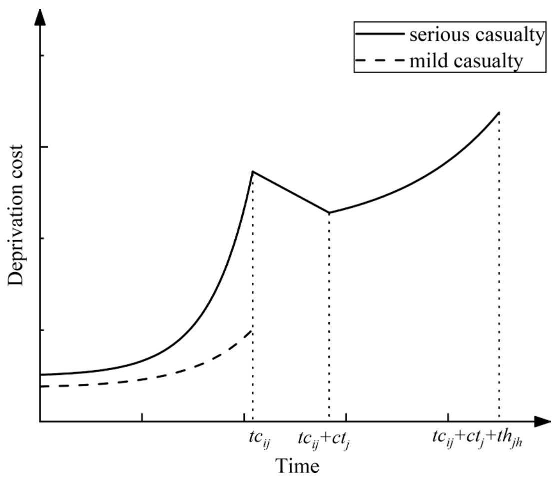


Fig. 2. The deprivation cost of the casualties.

(1) Temporary medical centers could be disrupted due to aftershocks. Once the temporary medical centers are disrupted, they are completely unavailable. At most two temporary medical centers are disrupted in each scenario, and the disruption of each temporary medical center is independent (Peng et al., 2011).
(2) Scenarios are divided according to the disrupted temporary medical centers and the historical data is considered sufficient to estimate the probability distribution. However, since the historical information is incomplete, the casualty number under each scenario is considered as uncertainty and is described as interval data (Li et al., 2020).
(3) The capacities of temporary medical centers and hospitals for the casualty are limited. And the hospitals have a limited supply of relief commodities.
(4) Mild and serious casualties need different amounts of relief commodities.

3.1. Mathematical model

In this sub-section, we formulate the mathematical model that is to be optimized, including the sets, parameters, decision variables, objective functions, and constraints.

3.1.1. Notations

Sets
I Set of disaster sites, $i \in I$
J Set of candidate temporary medical centers, $j \in J$
H Set of candidate hospitals, $h \in H$
K Set of relief commodities, $k \in K$
S Set of scenarios, $s \in S$
Parameters
$\widetilde{q}_i^s$ Number of mild casualties at disaster site i in scenario s (persons)
$\widetilde{n}_i^s$ Number of serious casualties at disaster site i in scenario s (persons)
c_j Capacity of temporary medical center j for casualties (persons)
w_{jk} Capacity of temporary medical center j for relief commodity k (units)
r_h Capacity of hospital h for casualties (persons)
v_{hk} Maximum supply of hospital h for relief commodity k (units)
tc_{ij} Transportation time from disaster site i to temporary medical center j (hours)
th_{jh} Transportation time from temporary medical center j to hospital h (hours)
f_j Fixed cost of establishing temporary medical center j (RMB)
tp_{ij} Unit transportation cost of casualty from disaster site i to temporary medical center j (RMB/person)
tf_{jh} Unit transportation cost of casualty from temporary medical center j to hospital h (RMB/person)
tt_{hjk} Unit transportation cost of relief commodity k from hospital h to temporary medical center j (RMB/unit)
ac_k Quantity of relief commodity k needed by a mild casualty (units)
aw_k Quantity of relief commodity k needed by a serious casualty (units)
ct_j Rapid treatment time for a serious casualty at temporary medical center j (hours)
φ_j^s Binary parameter which is equal 1 if the temporary medical center j is disrupted in scenario s, 0 otherwise
p_s Occurrence probability of scenario s
M A sufficient large number
ω Infeasibility penalty coefficient
α Minimum transported percentage of mild casualties
β Minimum transported percentage of serious casualties
$g_1, g_2, g_3, g_4, h_1, h_2$ Coefficient of the deprivation cost function
Auxiliary variables
ΓC_i^s The total deprivation cost of the transported mild casualties at disaster site i in scenario s
ΓW_i^s The total deprivation cost of the transported serious casualties at disaster site i in scenario s
Variables
Z_j 1 if the temporary medical center j is selected and 0 otherwise
T_h 1 if the hospital h is selected and 0 otherwise
X_{ij}^s Number of mild casualties transported from disaster site i to temporary medical center j in scenario s
Y_{ijh}^s Number of serious casualties transported from disaster site i to hospital h via temporary medical center j in scenario s
G_{hjk}^s Quantity of relief commodity k transported from hospital h to temporary medical center j in scenario s
δ_{is}^1 Number of mild casualties who are not transported at disaster site i in scenario s
δ_{is}^2 Number of serious casualties who are not transported at disaster site i in scenario s

3.1.2. Objective function formulation

Generally, the main goal of emergency relief problems is to reduce the suffering of casualties and transfer them to medical facilities for treatments as soon as possible. However, the casualties may not all be transported due to facility capacity and budget constraints. We consider a penalty for casualties who are not transferred in this model to deal with this situation. The penalty value is set large enough to make the available capacity of medical facilities be used, which is incorporated into the first objective function.

$$\min F1 = \sum_{s\in S} p_s \left[\sum_{i\in I} \Gamma C_i^s + \sum_{i\in I} \Gamma W_i^s + \omega \sum_{i\in I} \left(\delta_{is}^1 + \delta_{is}^2 \right) \right] \tag{1}$$

The objective function (1) minimizes the total deprivation cost for all casualties, which consists of the deprivation cost of the transferred casualties, and the penalty cost of un-transferred casualties.

$$\min F2 = \sum_{j\in J} f_j Z_j + \sum_{s\in S} P_s \left[\begin{array}{c} \sum_{i\in I}\sum_{j\in J} tp_{ij} X_{ij}^s + \sum_{i\in I}\sum_{j\in J}\sum_{h\in H} \left(tp_{ij} + tf_{jh} \right) Y_{ijh}^s + \\ \sum_{h\in H}\sum_{j\in J}\sum_{k\in K} tt_{hjk} G_{hjk}^s \end{array} \right] \tag{2}$$

The objective function (2) focuses on minimizing the total operation cost. The first term denotes the fixed cost of establishing temporary medical centers. The second term represents the cost of transporting mild casualties from disaster units to temporary medical centers. The third term is the transportation cost of serious casualties from disaster sites to hospitals via temporary medical centers. The fourth term calculates the transportation cost of all relief commodities.

3.1.3. Constraints

$$\sum_{j\in J} X_{ij}^s * \int_0^{tc_{ij}} \left(e^{g_1 t} + e^{h_1} \right) dt = \Gamma C_i^s, \forall i \in I, s \in S \tag{3}$$

$$\sum_{j\in J}\sum_{h\in H} Y_{ijh}^s * \left(\begin{array}{c} \int_0^{tc_{ij}} \left(e^{g_2 t} + e^{h_2} \right) dt + \int_{tc_{ij}}^{tc_{ij}+ct_j} \left(-g_3 t + h_3 \right) dt \\ + \int_{tc_{ij}+ct_j}^{tc_{ij}+ct_j+th_{jh}} \left(e^{g_4 t} + e^{h_4} \right) dt \end{array} \right) = \Gamma W_i^s, \forall i \in I, s \in S \tag{4}$$

$$\sum_{i\in I} X_{ij}^s + \sum_{i\in I}\sum_{h\in H} Y_{ijh}^s \le c_j, \forall j \in J, s \in S \tag{5}$$

$$\sum_{h\in H} G_{hjk}^s \le w_{jk}, \forall j \in J, k \in K, s \in S \tag{6}$$

$$\sum_{i\in I}\sum_{j\in J} Y_{ijh}^s \le r_h, \forall h \in H, s \in S \tag{7}$$

$$\sum_{j\in J} G_{hjk}^s \le v_{hk}, \forall h \in H, k \in K, s \in S \tag{8}$$

$$X_{ij}^s \le M^* Z_j \left(1 - \varphi_j^s \right), \forall i \in I, j \in J, s \in S \tag{9}$$

$$Y_{ijh}^s \le M^* Z_j \left(1 - \varphi_j^s \right), \forall i \in I, j \in J, h \in H, s \in S \tag{10}$$

$$Y_{ijh}^s \le M^* T_h, \forall i \in I, j \in J, h \in H, s \in S \tag{11}$$

$$G_{hjk}^s \le M^* T_h, \forall h \in H, j \in J, k \in K, s \in S \tag{12}$$

$$\sum_{j\in J} X_{ij}^s + \delta_{is}^1 \ge \widetilde{q}_i^s, \forall i \in I, s \in S \tag{13}$$

$$\sum_{j\in J}\sum_{h\in H} Y_{ijh}^s + \delta_{is}^2 \ge \widetilde{n}_i^s, \forall i \in I, s \in S \tag{14}$$

$$\sum_{h\in H} G_{hjk}^s \ge ac_k \sum_{i\in I} X_{ij}^s + aw_k \sum_{i\in I}\sum_{h\in H} Y_{ijh}^s, \forall j \in J, k \in K, s \in S \tag{15}$$

$$\sum_{j\in J} X_{ij}^s \ge \alpha^* \widetilde{q}_i^s, \forall i \in I, s \in S \tag{16}$$

$$\sum_{j\in J}\sum_{h\in H} Y_{ijh}^s \ge \beta^* \widetilde{n}_i^s, \forall i \in I, s \in S \tag{17}$$

$$Z_j, T_h \in \{0,1\}, \forall j \in J, h \in H \tag{18}$$

$$X_{ij}^{s}, Y_{ijh}^{s}, G_{hjk}^{s}, \delta_{is}^{1}, \delta_{is}^{2} \geq 0 \text{ and integer}, \forall i \in I, j \in J, h \in H, k \in K, s \in S \tag{19}$$

Constraints (3) and (4) respectively calculate the deprivation cost of the transferred mild casualty and the transferred serious casualty at each disaster site. Constraints (5) and (6) denote the capacity limitations of the temporary medical centers for casualties and relief commodities, respectively. Constraint (7) represents that the transferred casualties from all disaster sites do not exceed the capacity of hospital. Constraint (8) ensures that the supply for relief commodities of each hospital is not exceeded. According to constraints (9) and (10), only temporary medical centers that have been opened and are not disrupted can provide services. Constraints (11)–(12) prevent conflicts between decision variables. Constraints (13) and (14) signify that the casualties at each disaster site may not be fully transferred. Constraint (15) guarantees that the relief commodities transported from hospitals can meet the needs of casualties at each temporary medical center. Constraints (16) and (17) enforces that the number of transported casualties as a percentage of the total casualties at each disaster site is not less than a specified value. Constraints (18) and (19) define the types of all the decision variables.

3.2. Bi-objective model transformation using the ε-constraint method

Our proposed model includes two objective functions that have to be optimized under the same decision variables. Since the two objective functions are conflicting, there exists no solution that optimizes the objectives simultaneously. Therefore, we should adopt multi-objective solution procedures to obtain non-dominant solutions or Pareto solution sets that achieve a satisfactory balance between objectives.

There are various kinds of methods to solve the multi-objective problem, which can be roughly divided into interactive, decision-aided, meta-heuristic, scalar, and fuzzy methods. The frequently used methods for handling multi-objective problems are the ε-constraint, weighted sum, weighted metric, goal programming method, and lexicon method (Liu and Papageorgiou, 2013). Among these approaches, the weighted sum method is the most widely used (Balaman, 2016).

Compared to the weighted sum approach, the ε-constraint method has the following benefits: (1) In the ε-constraint method, the number of feasible solutions can be controlled through adjusting the preferred breakpoints in the range of each objective function. However, it is difficult to realize by using the weighted sum method because the solutions heavily rely on the specified weights (Mavrotas, 2009). (2) The weighted sum method only assists in generating extremely effective solutions. On the contrary, the ε-constraint method can also obtain non-extremely effective solutions (Rezvani et al., 2015). Many scholars have applied the ε-constraint method to deal with the multi-objective optimization problem (Haghi et al., 2017; Diabat et al., 2019; Hamdan and Diabat, 2020; Sun et al., 2021; Wattanasaeng and Ransikarbum, 2021). In this paper, the ε-constraint method proposed by Haimes et al. (1971) is applied to solve the proposed bi-objective model.

In the ε-constraint method, only one objective function is optimized while other objective functions are transformed into constraints. The general form of this method is shown in Eq. (20). Note that ε_i is determined by decision-makers and indicates the acceptable threshold of the objective $f_i(x)$.

$$\begin{aligned} & Min\, f_k(x) \\ & f_i(x) \leq \varepsilon_i, \forall i \neq k \\ & x \in X \end{aligned} \tag{20}$$

We use the ε-constraint method to solve the proposed model. The first objective function in Eq. (1) is selected as the main objective and the second objective function in Eq. (2) turns into a constraint. Therefore, the transformed single-objective model is as follows:

(1)

$$s.t. \sum_{j \in J} f_j Z_j + \sum_{s \in S} p_s \left[\begin{array}{c} \sum_{i \in I}\sum_{j \in J} tp_{ij} X_{ij}^{s} + \sum_{i \in I}\sum_{j \in J}\sum_{h \in H} \left(tp_{ij} + tf_{jh}\right) Y_{ijh}^{s} + \\ \sum_{h \in H}\sum_{j \in J}\sum_{k \in K} tt_{hjk} G_{hjk}^{s} \end{array} \right] \leq \varepsilon \tag{21}$$

$$constraints \quad (3) - (19)$$

3.3. The Interval robust optimization to handle uncertainty

In this paper, the number of mild casualties and serious casualties in each scenario is considered as uncertainties, which exists in constraints (13)–(14) and (16)–(17). We apply the robust method proposed by Ben-Tal et al. (2009) to cope with the uncertainties about the casualty number. According to their method, the number of mild casualties $\widetilde{q}_i^s$ and serious casualties $\widetilde{n}_i^s$ can be modeled as distribution-free parameters with the range of $\left[\overline{q}_i^s - \widehat{q}_i^s, \overline{q}_i^s + \widehat{q}_i^s\right]$ and $\left[\overline{n}_i^s - \widehat{n}_i^s, \overline{n}_i^s + \widehat{n}_i^s\right]$. Notation $\overline{q}_i^s(\overline{n}_i^s)$ is the mean of the interval and is named nominal value, while $\widehat{q}_i^s(\widehat{n}_i^s)$ represents the maximum deviation from $\overline{q}_i^s(\overline{n}_i^s)$. Then, $\Gamma_i^s \in [0,1]$ and $\Gamma'^s_i \in [0,1]$ are introduced to control the actual deviation, which is named the budget of uncertainty. Applying the uncertain budget to adjust the upper bound of the casualty number, the constraints (13)–(14) and (16)–(17) is reformulated as follows:

$$\sum_{j\in J}X_{ij}^{s}+\delta_{is}^{1}\geq\overline{q}_{i}^{s}+\Gamma_{i}^{s}\widehat{q}_{i}^{s},\forall i\in I,s\in S \tag{22}$$

$$\sum_{j\in J}\sum_{h\in H}Y_{ijh}^{s}+\delta_{is}^{2}\geq\overline{n}_{i}^{s}+\Gamma'^{s}_{i}\widehat{n}_{i}^{s},\forall i\in I,s\in S \tag{23}$$

$$\sum_{j\in J}X_{ij}^{s}\geq\alpha*\left(\overline{q}_{i}^{s}+\Gamma_{i}^{s}\widehat{q}_{i}^{s}\right),\forall i\in I,s\in S \tag{24}$$

$$\sum_{j\in J}\sum_{h\in H}Y_{ijh}^{s}\geq\beta*\left(\overline{n}_{i}^{s}+\Gamma'^{s}_{i}\widehat{n}_{i}^{s}\right),\forall i\in I,s\in S \tag{25}$$

The uncertain budget can be used to weigh the feasibility against the conservatism of the solutions. In particular, $\Gamma_i^s=0$ denotes the nominal situation, while $\Gamma_i^s=1$ represents the worst-case. Increasing Γ_i^s and Γ'^s_i will lead to a more conservative solution, because this model contains a greater degree of uncertainty. Decision-makers can set an appropriate value for Γ_i^s and Γ'^s_i based on their attitude to risk. In general, more risk-averse decision-makers tend to choose larger values.

Therefore, our proposed model can be reformulated as P1 model:

P1:

$$(1)$$

$$s.t.\quad(3)-(12),(15),(18)-(19),(21)-(25)$$

3.4. The novel scenario-based robust optimization

Mulvey et al. (1995) integrated data based on scenarios with goal programming and developed an approach called the scenario-based robust optimization. The scenario-based robust optimization method is less sensitive to changes of the data in a set of scenarios, because it can generate a stable and efficient solution for any possible realization of uncertain parameters. Moreover, it is the extension of the stochastic programming by introducing two types of robustness: (1) solution robustness means that the solution is close to optimal over all scenarios, and (2) model robustness refers to the fact that the solution is almost feasible when the input data changes. By introducing γ and ω, which refer to the weights of the two types of robustness, P1 model can be reformulated as follows:

$$\min FF1=\sum_{s\in S}p_sZ1_s+\gamma\sum_{s\in S}p_s\Big(Z1_s-\sum_{s\in S}p_sZ1_s\Big)^2+\omega\sum_{s\in S}\sum_{i\in I}p_s\left(\delta_{is}^1+\delta_{is}^2\right) \tag{26}$$

$$s.t.\ Z1_s=\sum_{i\in I}\Gamma C_i^s+\sum_{i\in I}\Gamma W_i^s,\forall s\in S \tag{27}$$

$$constrints\ (3)-(12),(15),(18)-(19),(21)-(25)$$

In the objective function, the first term represents the expected value, the second term indicates solution robustness which is measured by the variance of objective value for each scenario to deal with risk, and the third term signifies model robustness which is used to penalize infeasibility. This model focuses on making a trade-off between the two kinds of robustness by changing weight ω. There is a quadratic term that exists in Eq. (26), which increases the difficulty of solving models. According to Yu and Li (2000), Eq. (26) can be rewritten by:

$$\min FF1=\sum_{s\in S}p_sZ1_s+\gamma\sum_{s\in S}p_s\Big|Z1_s-\sum_{s\in S}p_sZ1_s\Big|+\omega\sum_{s\in S}\sum_{i\in I}p_s\left(\delta_{is}^1+\delta_{is}^2\right) \tag{28}$$

Eq. (28) remains nonlinear, which makes the model difficult to solve. To convert Eq. (28) to a linear form, Leung et al. (2007) introduced an effective linearization method based on Li (1996), which can reduce the number of non-negative variables. Then, Eq. (28) is linearized by introducing $\theta 1_s$, and our NSRB optimization model can be formulated as follows using this method (see P2 model):

P2:

$$\min FF1=\sum_{s\in S}p_sZ1_s+\gamma\sum_{s\in S}p_s\Big(Z1_s-\sum_{s\in S}p_sZ1_s+2\theta1_s\Big)+\omega\sum_{s\in S}\sum_{i\in I}p_s\left(\delta_{is}^1+\delta_{is}^2\right) \tag{29}$$

$$s.t.\ Z1_s-\sum_{s\in S}p_sZ1_s+\theta1_s\geq0,\forall s\in S \tag{30}$$

$$\theta1_s\geq0,\forall s\in S \tag{31}$$

$$constrints\ (3)-(12),(15),(18)-(19),(21)-(25),(27)$$

Since the objective function aims to achieve the minimum value, it is clear that $\theta_s=0$ if $Z1_s-\sum_{s\in S}P_sZ1_s\geq0$. In this case, the objective function becomes $\sum_{s\in S}P_sZ1_s+\gamma\sum_{s\in S}P_s\left(Z1_s-\sum_{s\in S}P_sZ1_s\right)+\omega\sum_{s\in S}\sum_{i\in I}p_s\left(\delta_{is}^1+\delta_{is}^2\right)$. Besides, if $Z1_s-\sum_{s\in S}P_sZ1_s<0$, the objective function will be optimized when $\theta_s = \sum_{s\in S} P_s Z1_s - Z1_s$. Then the objective function turns into $\sum_{s\in S} P_s Z1_s + \gamma \sum_{s\in S} P_s \left(\sum_{s\in S} P_s Z1_s - Z1_s\right) + \omega \sum_{s\in S} \sum_{i\in I} p_s \left(\delta^1_{is} + \delta^2_{is}\right)$.

4. Computational experiments

In this section, several numerical experiments are conducted to test the proposed NSRB model. We consider the real case of the Wenchuan Earthquake in 2008 using the data from the China Earthquake Administration. The model is solved by IBM ILOG CPLEX 12.6.1 on the PC with 2.11 GHz Intel i5 CPU and 8 GB RAM.

4.1. Background

On May 12, 2008, an 8.0 magnitude earthquake happened in Wenchuan City, Sichuan Province. This earthquake resulted in a large number of aftershocks and secondary disasters. Many people were injured and left homeless in this devastating disaster. As shown in Fig. 3, 10 severely affected cities or towns are selected as disaster sites. There are both mild and serious casualties at each disaster site. And we considered 10 candidate temporary medical centers in the cities or towns with less damage on roads and buildings. Besides, 4 medical facilities with relatively good conditions that are far from the epicenter are selected as candidate hospitals. I1-I10 represent the disaster sites, J1-J10 indicate the candidate temporary medical centers, and H1-H4 denote the candidate hospitals. Based on the historical data, J2, J4 and J7 may be affected by aftershocks and disrupted. To evaluate our proposed model, 7 different scenarios are generated based on the recommendations of the experts and historical data. In particular, the first scenario represents the situation where there is no disruption of temporary medical centers. In other scenarios, one or two temporary medical centers are out of service.

Based on the historical information, locally published reports, and related research (Liu et al., 2019; Ni et al.,2018), Table 2 gives the nominal values of the casualty number for 10 disaster sites when there is no disruption. The travel time between disaster sites and temporary medical centers is reported in Table 3, and the travel time between temporary medical centers and hospitals is shown in Table 4. The transportation cost is approximately estimated based on the actual distance and the cost of labor and fuel (Ni et al., 2018). The casualty capacity of each temporary medical center and hospital is set at 700 and 600, respectively. In this paper, the two main types of supplies needed by the casualty are considered: medical equipment and medicine. The demands of a mild casualty for medical equipment and medicine are 2 units and 1 unit. A serious casualty needs 1 unit of medical equipment and 3 units of medicine (Alem

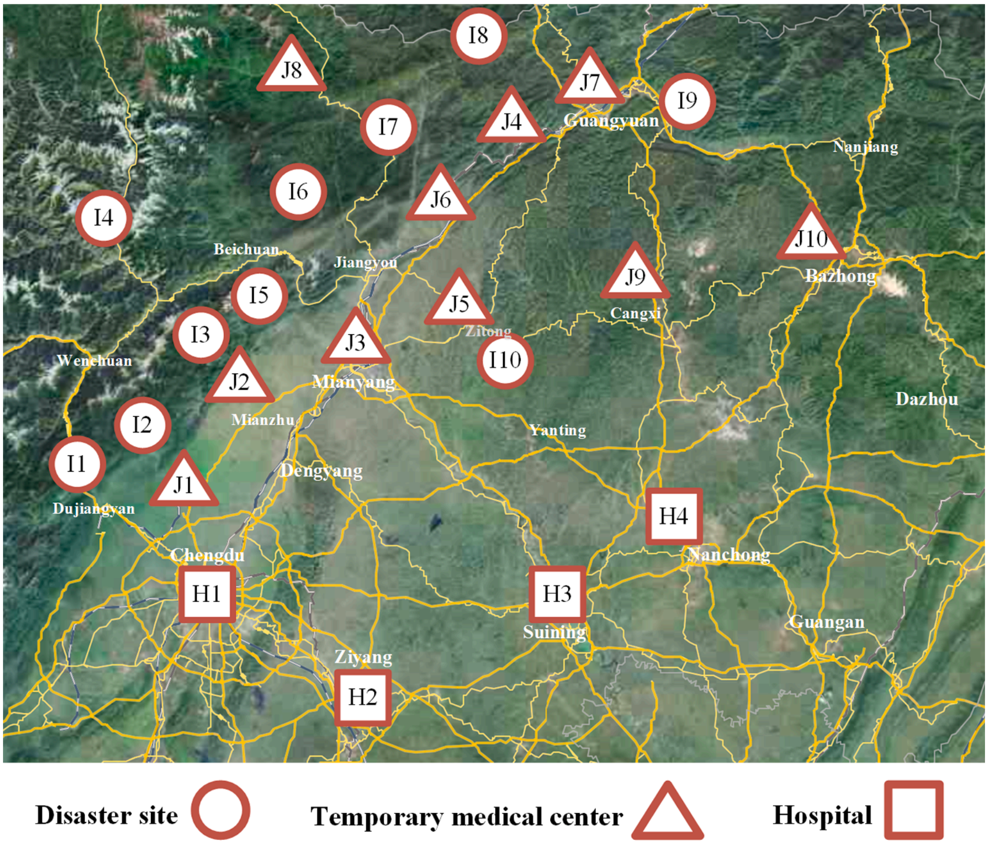


Fig. 3. Map of disaster sites, candidate temporary medical centers, and candidate hospitals.

Table 2
Nominal values of casualty number for disaster sites under no disruptions (persons).

Disaster sites	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10
Mild casualty	478	65	292	97	438	406	450	182	78	63
Serious casualty	287	35	158	53	262	244	270	98	42	34

Table 3
Transportation time between disaster sites and candidate temporary medical centers (hours).

Disaster sites	Candidate temporary medical centers									
	J1	J2	J3	J4	J5	J6	J7	J8	J9	J10
I1	1.17	2.00	3.00	4.67	3.88	3.97	5.43	4.95	6.07	6.38
I2	1.07	1.25	2.30	3.95	3.18	3.27	4.73	4.80	5.47	6.28
I3	1.00	0.50	1.37	3.00	2.08	2.30	3.80	3.37	5.53	6.13
I4	3.00	2.83	2.78	4.50	3.37	3.17	5.27	3.65	6.17	6.20
I5	1.35	0.50	0.83	2.70	1.57	1.78	3.30	2.68	4.20	5.12
I6	2.42	1.50	1.50	2.22	1.62	1.30	3.00	1.62	4.22	5.35
I7	3.18	2.27	1.85	1.90	2.05	1.57	3.00	1.23	3.93	5.80
I8	4.05	3.83	2.92	1.00	2.87	1.73	1.77	2.22	2.85	4.12
I9	4.17	4.45	3.55	1.95	3.17	2.38	1.20	3.98	4.22	4.60
I10	3.00	2.33	1.18	2.60	0.98	1.68	3.15	4.58	3.75	3.77

Table 4
Transportation time between candidate temporary medical centers and candidate hospitals (hours).

Candidate temporary medical centers	Candidate hospitals			
	H1	H2	H3	H4
J1	1.02	2.32	3.22	4.20
J2	1.52	2.80	3.55	3.72
J3	2.47	3.43	2.58	2.73
J4	4.13	5.10	4.37	4.23
J5	3.20	4.35	3.35	2.77
J6	3.25	4.42	3.73	3.88
J7	4.92	5.88	4.62	3.78
J8	4.63	5.85	5.25	5.42
J9	4.72	5.78	4.82	4.85
J10	5.63	5.68	3.73	4.80

et al.,2016). Each temporary medical center has adequate capacity for 2,000 units of medical equipment and 3,000 units of medicine. The maximum supply of each hospital for the two types of relief commodities is 2300 and 2500 units, respectively. The fixed cost for temporary medical centers is related to capacity and is set to 150,000 RMB (Üster and Dalal, 2017). To ensure that the percentage of transported casualties exceeds a specific value, α and β are both considered as 0.7 (Habibi-Kouchaksaraei et al., 2018). In the deprivation cost function, $g_1 = 0.4, g_2 = 1.2, g_3 = 20, g_4 = 0.5, h_1 = 3.2, h_2 = 3.5$ based on Zhu et al (2019b). To ensure that the deprivation cost for serious casualties is continuous, the parameters h_3 and h_4 are not fixed in advance, but are adjusted according to

Table 5
Optimal solution with different.ε_2

NO.	ε_2	$FF1$	Weighted un-transferred casualties
1	1.50×10^6	12365415.69	12,141,520
2	1.60×10^6	9925228.98	9,690,980
3	1.70×10^6	5800479.35	5,537,440
4	1.80×10^6	4077871.27	3,807,540
5	1.90×10^6	2797192.02	2,470,140
6	1.95×10^6	1354654.10	1,006,680
7	1.98×10^6	557745.78	192,760
8	2.00×10^6	349670.32	0
9	2.10×10^6	323928.96	0
10	2.20×10^6	289982.21	0
11	2.30×10^6	289982.21	0

the values of the function. Besides, the value of λ is set to 1(Safaei et al., 2018). Through several numerical experiments, we find that the certain penalty value may no longer work as the uncertain level of casualty number increases. In fact, the higher uncertain level of casualty number means that more casualties need to be transported after earthquakes. As the number of casualties increases, the deprivation cost of some casualties may increase. Then the certain penalty value may be less than the deprivation cost of some casualties. As a result, some casualties cannot be transferred even if the resources (facility capacity, supply, and cost) are sufficient. This phenomenon is also verified by Liu et al. (2018). To make sure that the penalty coefficient works well in all cases, we set $\omega = 20000$, which is greater than the maximum possible deprivation cost in the case studies.

4.2. Computation results under nominal values

To explore the relationship between the two objectives, we first solve the P2 model using nominal values without the consideration of uncertain casualty number. By solving each objective function individually, the optimal values of the main objective and the secondary objective are 289982.21 and 1392486.13, respectively. We utilize the hierarchical sequence approach to calibrate the parameters, $FF1^* = 289982.21$, $\varepsilon_2 \geq 1392486.13$. In this work, we consider 11 different values for ε_2. Table 5 summarizes the results of the main objective function with different values of ε_2. In the fourth column, the weighted un-transferred casualties refers to $\omega\sum_{s\in S}\sum_{i\in I}p_s\left(\delta_{is}^1+\delta_{is}^2\right)$. More specifically, Fig. 4 shows the Pareto front for the two objectives by appropriately selecting ε_2. As can be seen from Table 5, the increasing operation cost leads to a decrease in the deprivation cost of casualty and the number of un-transferred casualties. When the value of ε_2 reaches 2.20×10^6, the main objective function value achieves its minimum value. However, when ε_2 is greater than 2.20×10^6, the optimal value of FF_1 no longer changes as ε_2 further increases. This phenomenon indicates a contradiction of benefits between the two objective functions. Decision-makers cannot find a solution that is optimal for both objectives at the same time. Thus, they should make a trade-off between humanitarianism and cost by setting an appropriate value for ε_2 according to their preferences. For example, managers who are more concerned about humanitarianism may be willing to pay 2.20×10^6 instead of 2.00×10^6 RMB to decrease the deprivation cost of all casualties. However, with the assurance that all casualties can be transported, managers who prefer to save costs may choose to pay 2.00×10^6 rather than 2.20×10^6 RMB.

The best solution under the scenario with no disruption when $\varepsilon_2 = 2.20 \times 10^6$ is shown in Fig. 5. According to the decision results, 8 temporary facilities and 4 hospitals are opened to meet the needs of casualty evacuation. There are two values in each parenthesis near dotted lines between disaster sites and temporary medical centers indicating the number of transported mild and serious casualties. The values on the dotted lines represent the number of serious casualties transported from temporary medical centers to hospitals. The lines from hospitals to temporary medical centers refer to the flows of relief commodities. Each square bracket near lines contains two values, which are the amount of transported medical equipment and medicine. Fig. 5 shows that the relief campaigns are performed at relatively close distances to effectively rescue the casualties. In this case, since our proposed model focus on minimizing the deprivation cost, only some of the medical facilities are selected and all casualties are transferred because of sufficient resources (facility capacity, supply, and cost). While resources are inadequate, all medical facilities may be opened and a few casualties cannot be transported to medical facilities for treatment. Those un-transferred casualties are given a penalty value in the main objective function.

4.3. Solution robustness analysis

In this subsection, the solution robustness of our proposed model is evaluated by the comparison with the hybrid robust and stochastic bi-objective (HRSB) model when $\Gamma_i^s = \Gamma'^s_i = 0.2$ and data variability = 15%. According to the description above, when γ equals 0, our proposed model (i.e., P2 model) then becomes the HRSB model (i.e., P1 model). In the HRSB model, only the interval robust method is applied, and the objective function only focuses on minimizing the deprivation cost of transported casualties and the penalized deprivation cost of un-transferred casualties. However, it is not enough to have a solution with minimum expected

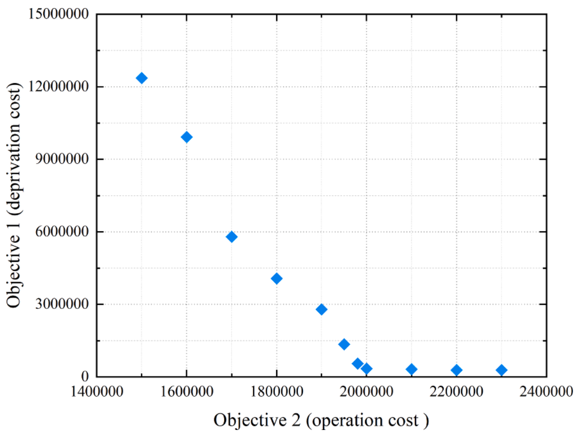


Fig. 4. Pareto front of the two objective functions.

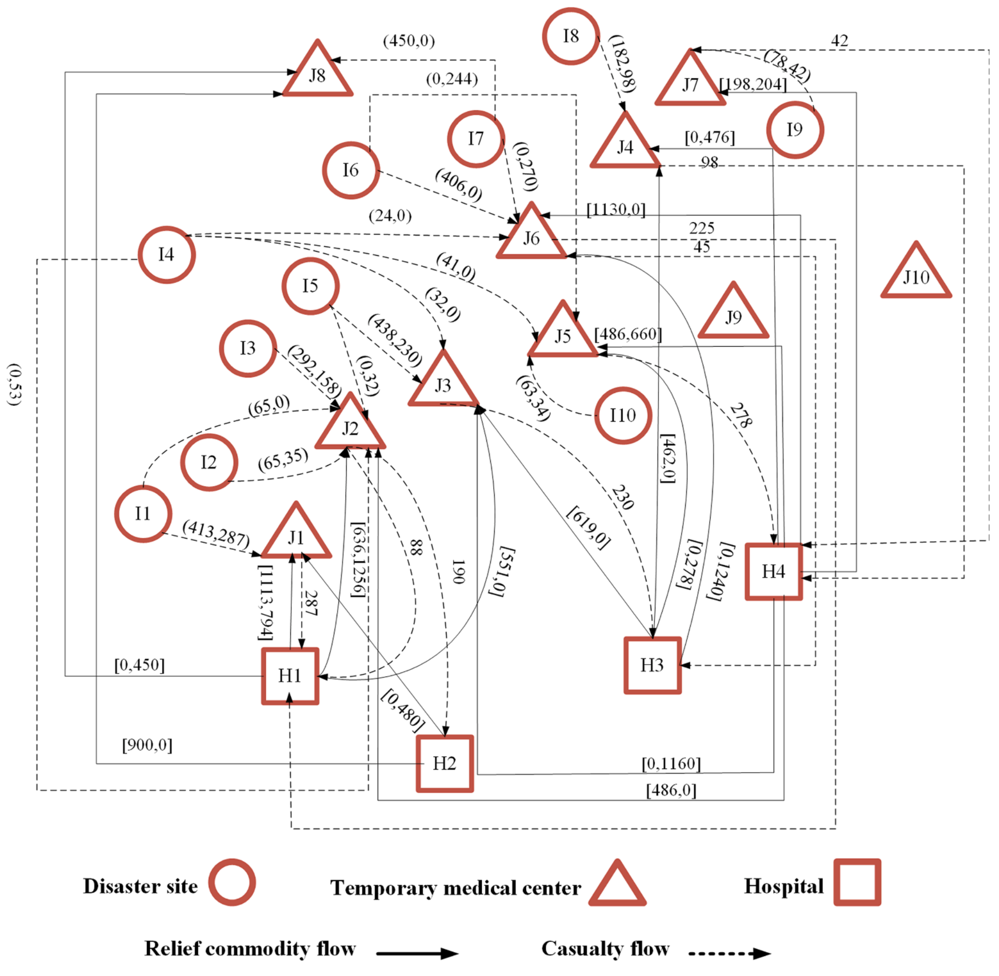


Fig. 5. Decision results under nominal values.

deprivation cost during the actual rescue process, the solution also needs to have low fluctuations under various disruption scenarios. The main results of the two approaches with $\varepsilon_2 = 2.20 \times 10^6$ are reported in Table 6. The expected deprivation cost is calculated by $\sum_{s\in S} p_s Z1_s$, and the variance of deprivation cost refers to $\sum_{s\in S} p_s \left(Z1_s - \sum_{s\in S} p_s Z1_s + 2\theta 1_s\right)$. As can be seen, the expected deprivation cost of the NSRB model is 0.89% higher than that of the HRSB model. In terms of the variance of deprivation cost, the NSRB model is 65.02% lower than the HRSB model. This reveals that the solution of our model is significantly more stable across all scenarios, despite a slight increase in the expected deprivation cost.

More specifically, Fig. 6 illustrates the deprivation cost under each scenario. We can see that the solution generated by our proposed model is less sensitive against various scenarios, because the curve drawn by the dotted line is less volatile. This verifies that our model is robust under different disruption scenarios. Since the robustness of the solution is essential for managers, our proposed model can be effectively used for emergency rescue networks to address the risk of facility disruptions.

4.4. Effect of considering the uncertainty

To investigate the impact of uncertain parameters on the main objective function value, we conduct some numerical experiments for the NSRB model. In the numerical experiments, the uncertain budget takes four different values in [0,1], and $\varepsilon_2 = 2.80 \times 10^6$. Under each specific budget of uncertainty, the maximum deviations of uncertainties are considered to be 5%, 10%, 15% or 20% of their

Table 6
Comparison of the NSRB and HRSB model.

	Expected deprivation cost	Variance of deprivation cost	Un-transferred casualties
NSRB	377826.24	2802.76	0
HRSB	374447.84	8011.34	0

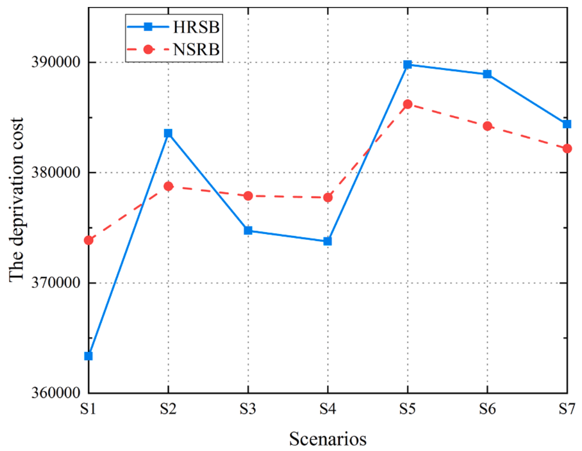


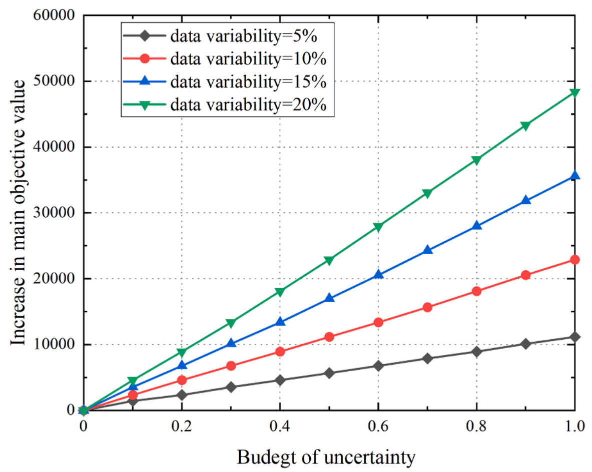
Fig. 6. The deprivation cost for each scenario.

nominal values. In other words, data variability is 5%, 10%, 15% or 20%. Since the number of casualties must be a positive integer, the real value of the casualty number is rounded up.

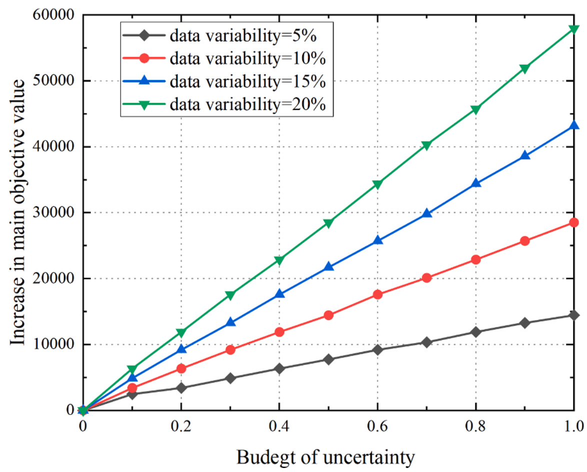
Fig. 7(a) and (b) depict the increase of the main objective values resulting from variations of serious casualty number and mild casualty number, respectively. As expected, it can be seen from Fig. 7(a) and (b) that the main objective function value increases linearly with the uncertain budget under each data variability. In particular, the worst values of the main objective function are achieved as the uncertain budget takes the maximum value. For each given uncertain budget, higher data variability results in a larger increase of the main objective function value. These results indicate that higher uncertain budget or data variability of casualty number has a greater influence on the main objective values.

By comparing Fig. 7(a) and (b), the uncertainty about the serious casualty number tends to have more effects on the main objective value than the mild casualty number. For instance, when the uncertainty budget and the data variability are 0.4 and 15%, respectively, the uncertainty in the number of mild casualties leads to an increase in the main objective function value by 13386.34. But the deterioration effect of the uncertain serious casualty number on the main objective function is 17590.92, which is greater than the former. These findings reveal excellent management insights that we should put more effort into accurately forecasting the number of serious casualties rather than mild casualties.

Next, we compare the performance of the NSRB model with the scenario-based robust bi-objective (SRB) model to study the effect of considering uncertainties. In the SRB model, only the scenario-based robust approach is utilized and the uncertain number of casualties is not considered. To be more convincing, we consider two other problems of different sizes. Firstly, the SRB model is solved using the nominal values, while our proposed model is solved separately under different uncertain budgets and data variability. Then we uniformly generate 50 realizations of uncertain casualty number within the corresponding interval range. For the solutions obtained at the beginning, we calculate the main objective function value under each realization of the uncertain data. Finally, the



(a) uncertainty about mild casualty number



(b) uncertainty about serious casualty number

Fig. 7. Sensitivity of uncertainty about mild casualty number and serious casualty number.

average value of the main objective among 50 realizations is calculated. The results of the comparison between the two models are reported in Table 7. From the results in columns 4 and 5, we can find that the main objective value solved by the SRB model is lower than the NSRB model regarding nominal values. However, it can be seen from columns 6 and 7 that the average value of the main objective obtained by our proposed model is much lower than the SRB model when applying 50 realizations. The main reason for this result is that the solution obtained by the SRB model is barely adaptable to all data realizations and some casualties cannot be transported, which can be verified by columns 8 and 9. On the contrary, the solutions obtained by our model can satisfy the demands of the casualty in all implementations.

4.5. Sensitivity analysis of changing capacities

In this subsection, numerical experiments are conducted to examine the main objective function sensitivity to changing the casualty capacities of temporary facilities and hospitals under the smallest test instance. We fix other parameters and change only the casualty capacities of medical facilities, the results of the main objective function value under four different uncertain levels are shown in Fig. 8. Under the four circumstances, the value of ε_2 is taken as 2.80×10^6. As can be seen from Fig. 8(a)-(d), when the variation in data or uncertain budget increases, changes in the casualty capacity of temporary medical centers and hospitals have a greater effect on the main objective value. As expected, an increase in casualty capacity of either temporary medical centers or hospitals can reduce the deprivation cost. According to Fig. 8(a)-(d), another important finding is that increasing the casualty capacity of temporary medical centers has greater impacts on the main objective value than hospitals. The reason may be that the temporary medical centers can receive both types of casualties, while the hospital treats only the serious casualties. Therefore, to improve the efficiency of rescue operations, the casualty capacity of temporary medical centers should be expanded first instead of hospitals when the resources are limited.

4.6. Performance considering facility disruptions or not

One of the main features of our work is considering facility disruptions, which distinguishes our work from most other works. Under the smallest test problem, we compare the solutions when considering facility disruptions or not. For the given uncertainty budget and data variability, we solve the model with/without the consideration of facility disruptions, respectively. Then, 50 data realizations of uncertain casualty number are uniformly generated in the corresponding interval. We use the solutions generated previously to obtain the corresponding main objective function value for each data realization. Then the mean objective function value among 50 realizations is calculated. All test instances are performed with $\varepsilon_2 = 2.80 \times 10^6$.

Table 8 summarizes the numerical results, where "Y" indicates that the facility disruption is considered, "N" means that the facility

Table 7
Comparison of the NSRB model versus the SRB model.

Problem size |I|*|J|*|H|*|S|	Budget of uncertainty	Data variability (%)	Main objective value under nominal data		Average value of the main objective		Average of the weighted un-transferred casualties	
			SRBB	NSRB	SRB	NSRB	SRB	NSRB
10*10*4*7	0.3	5	289982.21	298557.35	410893.59	294379.67	120900.00	0
($\varepsilon_2 = 2800000$)		15		313791.21	611116.26	301657.08	318760.00	0
		20		321955.15	788078.79	305544.22	495560.00	0
	0.7	5		308393.57	575379.09	299530.65	284440.00	0
		15		346570.83	1080181.57	315346.30	783380.00	0
		20		366652.65	1449206.16	327364.81	1147120.00	0
	1	5		316152.69	685063.32	302938.92	392860.00	0
		15		372199.48	1607674.53	329327.42	1305720.00	0
		20		402378.83	1956285.22	344579.26	1643980.00	0
15*13*5*11	0.3	5	340563.58	351941.09	446986.16	345971.43	103511.20	0
($\varepsilon_2 = 3800000$)		15		373713.40	891160.12	356232.75	546800.80	0
		20		385104.36	1087059.69	359704.07	742664.00	0
	0.7	5		365849.72	719925.50	352414.94	376142.00	0
		15		421780.11	1774074.91	375763.54	1424144.80	0
		20		452126.61	2203388.79	382948.00	1852234.80	0
	1	5		377163.68	951675.20	357215.05	607406.80	0
		15		460776.81	2442813.99	387794.99	2089848.80	0
		20		507270.14	2953620.68	400489.96	2594514.00	0
20*16*6*16	0.3	5	396230.90	409113.07	469690.90	404269.30	67010.40	0
($\varepsilon_2 = 4800000$)		15		437586.52	983840.00	414989.75	580953.60	0
		20		453044.30	1390716.64	421590.80	988477.20	0
	0.7	5		427381.32	734144.45	330919.20	410885.25	0
		15		501350.45	2235084.55	433995.76	1831273.20	0
		20		542823.27	2619116.01	443605.86	2212524.00	0
	1	5		442468.13	1094109.74	416688.75	691948.80	0
		15		555014.93	3127171.26	452918.64	2717463.60	0
		20		620008.56	4222971.74	470673.76	3806481.00	0

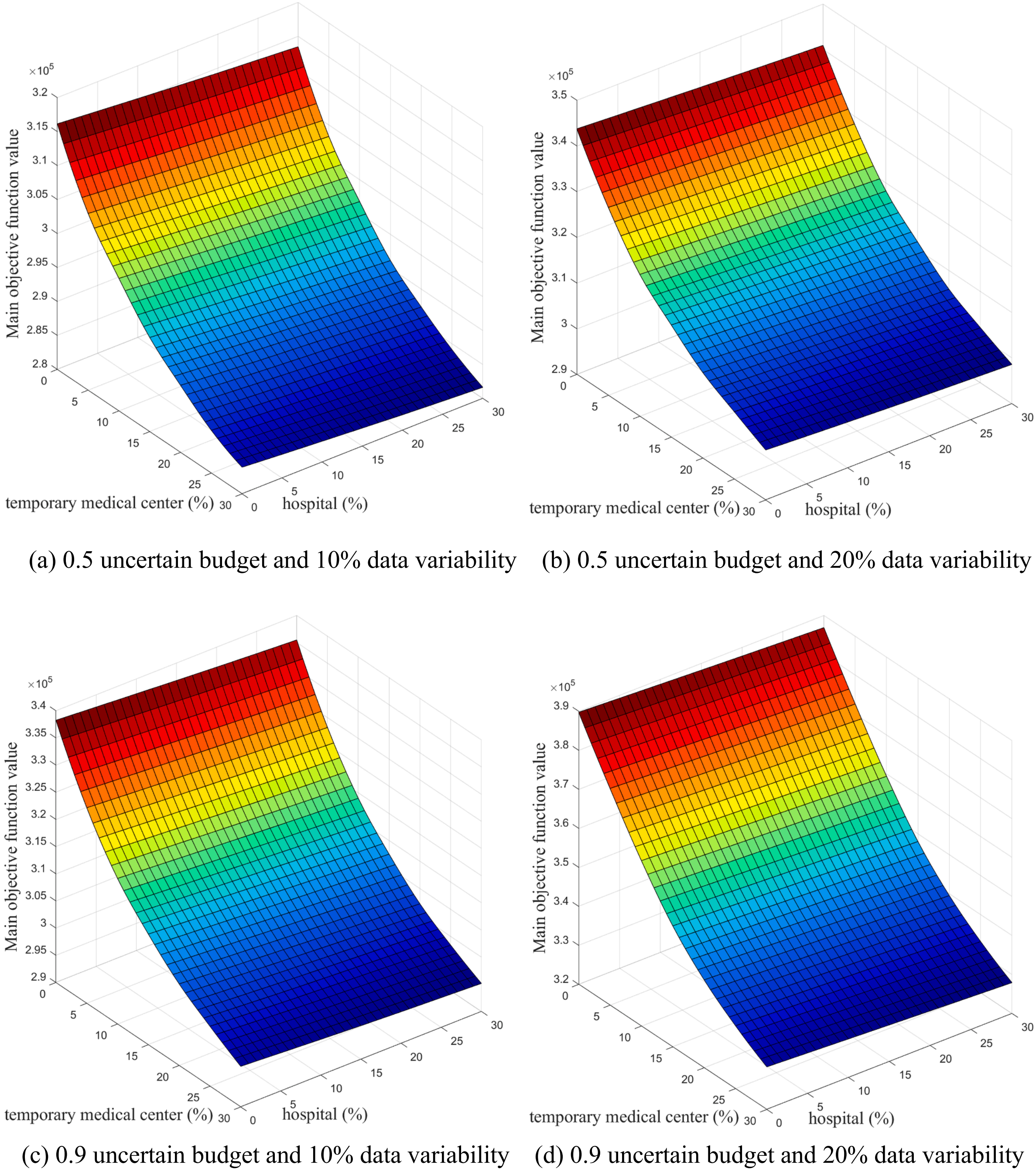


Fig. 8. Changes in the main objective value for increasing casualty capacity.

disruption is not considered. According to columns 3 and 4, under each setting for the uncertainty budget and data variability, the main objective function value is greater when considering facility disruptions. This may be because alternative facilities need to be selected to meet demands when some critical temporary medical centers are unavailable in certain scenarios, which increases the transportation time. However, we can see that the solutions can make all casualties transported to medical facilities when considering the disruptions of temporary medical centers. When the facility disruptions are not considered, there are some un-transferred casualties. Therefore, we can find that the solutions considering facility disruptions perform better in the humanitarian relief logistics.

5. Managerial insights

According to the results of computational experiments, some significant managerial implications are derived in support of rescue

Table 8
Performance considering facility disruptions or not.

Budget of uncertainty	Data variability (%)	Main objective valueunder nominal data		Average value of the main objective		Average of the weighted un-transferred casualties	
		Y	N	Y	N	Y	N
0.2	5	295863.72	259633.20	293032.28	354738.28	0	64220.00
	15	306124.05	266978.33	297958.85	533025.36	0	242060.00
	20	311072.84	270582.42	302236.47	661257.33	0	369000.00
0.4	5	301073.79	263426.23	295882.46	452074.82	0	161720.00
	15	321955.15	277790.90	306207.67	790759.65	0	497640.00
	20	332596.29	284837.88	310446.65	922906.20	0	627900.00
0.6	5	306124.05	266978.33	298299.15	533689.77	0	243100.00
	15	338259.55	288406.97	312609.94	994768.44	0	699140.00
	20	355254.67	299817.48	321285.49	1280037.62	0	981240.00
0.8	5	311072.84	270582.42	302910.26	662382.46	0	369980.00
	15	355254.67	299817.48	321993.25	1281368.92	0	980120.00
	20	377537.31	316676.14	331448.75	1629871.69	0	1327300.00
1	5	316152.69	273931.09	304143.82	744878.12	0	452920.00
	15	372199.48	312563.21	328960.74	1536840.81	0	1234350.00
	20	402378.83	334311.04	344580.95	2008170.10	0	1695200.00

operations. Firstly, the higher uncertain level of parameters can lead to the greater deprivation cost. Compared to the number of mild casualties, the number of serious casualties has a greater influence on the deprivation cost. Consequently, we can get a practical application that the decision-makers should focus on accurately predicting the number of serious casualties to decrease the total deprivation cost. Secondly, increasing the capacity of medical facilities can effectively reduce the deprivation cost for all casualties. In particular, increasing the casualty capacity of temporary medical centers has greater impacts on the deprivation cost than that of hospitals. Therefore, the managers should prioritize expanding the casualty capacity of temporary medical centers to decrease the deprivation cost for all casualties. Thirdly, the results show that the NSRB model can generate stable and feasible solutions across all disruption scenarios. Finally, considering both uncertainties and facility disruptions can save more casualties and improve the efficiency of rescue in real-world applications.

6. Conclusion

In this paper, a NSRB model is presented to design the HLN under uncertainty in casualty number. The decisions include determining the locations of the temporary medical centers and the hospitals, the plans of casualty transportation, and the strategies of relief commodity allocation under each disruption scenario. The rescue chain consists of disaster sites, temporary medical centers, and hospitals. The temporary medical centers are at risk of disruption. A set of scenarios are considered to model the disruption situations, and the robust interval set is used to handle the uncertain casualty number. Moreover, two different classes of casualties and the deprivation cost related to time are also considered in this model. The bi-objective framework is utilized to minimize the total deprivation cost and the total operation cost. The case studies based on the Wenchuan earthquake illustrate that our proposed model has better performance compared to other presented methods. According to the sensitivity analysis, we can find that the changes in the number of serious casualties have greater impacts on the main objective than mild casualties. Besides, expanding the capacity of temporary medical centers can more effectively reduce the deprivation cost of all casualties. On the other hand, the opened temporary medical centers and hospitals can satisfy more casualties when considering the disruptions and uncertainties. This verifies that the solution generated by considering the disruptions and uncertainties has better performance for data realizations.

The proposed model considers that only the temporary medical centers can be out of service, but the roads may also be disrupted in the real world. In future research, the disruption of routes can be investigated. As the rescue activities may last for a long time, we can focus on designing a multi-period integrated HLN to make real-time response plans. Besides, the computation time increases significantly with the increasing dimension of the problem. Another direction of future research could pay attention to developing an exact or heuristic approach to solve the large-scale problem.

CRediT authorship contribution statement

Huali Sun: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing, Funding acquisition, Project administration, Supervision. **Jiamei Li:** Conceptualization, Methodology, Software, Validation, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Tingsong Wang:** Methodology, Resources, Visualization, Formal analysis, Writing – review & editing. **Yaofeng Xue:** Writing – review & editing, Visualization, Formal analysis, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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